The documentation and process conversion measures necessary to comply with this revision shall be completed by 24 July 2025.

INCH-POUND

MIL-PRF-19500/706C
w/AMENDMENT 2
24 March 2005
SUPERSEDING
MIL-PRF-19500/706C
w/AMENDMENT 1
14 May 2020

PERFORMANCE SPECIFICATION SHEET

# TRANSISTOR, FIELD EFFECT, RADIATION HARDENED N-CHANNEL, SILICON, TYPES 2N7497T2, 2N7498T2, 2N7499T2, AND 2N7561T2 JANTXVR AND JANSR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

## 1. SCOPE

1.1 Scope. This specification covers the performance requirements for N-Channel, enhancement-mode, MOSFET, radiation hardened (total dose and single event effects (SEE)), power transistors, with avalanche energy maximum rating ($E_{AS}$) and maximum avalanche current ($I_{AS}$) for use in particular power-switching applications. Two levels of product assurance (JANTXV and JANS) are provided for each encapsulated device type.

1.2 Package outlines. The device package outlines are as follows: TO-205AF in accordance with figure 1 for all encapsulated device types.

1.3 Maximum ratings. $T_A$ = +25°C, unless otherwise specified.

| Type | $P_T$ (1) $T_C$ = +25°C | $P_T$ $T_A$ = +25°C | $R_{\theta JC}$ (2) | $V_{DS}$ | $V_{DG}$ | $V_{GS}$ | $I_{D1}$ (3) (4) $T_C$ =+25°C | $I_{D2}$ (3) (4) $T_C$ = +100°C | $I_S$ | $I_{DM}$ (5) | $T_J$ and $T_{STG}$ | $V_{ISO}$ 70,000 ft. altitude |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | W | W | °C/W | V dc | V dc | V dc | A dc | A dc | A dc | A (pk) | °C | V dc |
| 2N7497T2 | 25 | 0.71 | 5.0 | 130 | 130 | ±20 | 10.5 | 6.5 | 10.5 | 42 | -55 | N/A |
| 2N7498T2 | 25 | 0.71 | 5.0 | 200 | 200 | ±20 | 6.7 | 4.3 | 6.7 | 26.8 | to | N/A |
| 2N7499T2 | 25 | 0.71 | 5.0 | 250 | 250 | ±20 | 5.2 | 3.3 | 5.2 | 20.8 | +150 | 250 |
| 2N7561T2 | 25 | 0.71 | 5.0 | 250 | 250 | ±20 | 5.2 | 3.3 | 5.2 | 20.8 | | 250 |

(1) Derate linearly by 0.2 W/°C for $T_C$ > +25°C.
(2) See figure 2, thermal impedance curves.
(3) The following formula derives the maximum theoretical $I_D$ limit. $I_D$ is limited by package and device construction to 12 A.

$$I_D = \sqrt{\frac{T_{JM} - T_C}{(R_{\theta JC}) \times (R_{DS}(on) \text{ at } T_{JM})}}$$

(4) See figure 3, maximum drain current graphs.
* (5) $I_{DM}$ = 4 X $I_{D1}$ as defined in note (3).

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.4 Primary electrical characteristics at $T_C$ = +25°C.

| Type | Min $V_{(BR)DSS}$ $V_{GS}$ = 0 $I_D$ = 1.0 mA dc | $V_{GS(TH)1}$ $V_{DS} \geq V_{GS}$ $I_D$ = 1.0 mA dc | | Max $I_{DSS1}$ $V_{GS}$ = 0 $V_{DS}$ = 80 percent of rated $V_{DS}$ | Max $r_{DS(on)}$ (1) $V_{GS}$ = 12V, $I_D$ = $I_{D2}$ | | $E_{AS}$ |
|---|---|---|---|---|---|---|---|
| | | | | | $T_J$ = +25°C | $T_J$ = +150°C | |
| | V dc | V dc | | µA dc | Ω | Ω | mJ |
| | | Min | Max | | | | |
| 2N7497T2 | 130 | 2.5 | 4.5 | 10 | 0.100 | 0.230 | 164 |
| 2N7498T2 | 200 | 2.5 | 4.5 | 10 | 0.240 | 0.552 | 149 |
| 2N7499T2 | 250 | 2.5 | 4.5 | 10 | 0.420 | 0.924 | 142 |
| 2N7561T2 | 250 | 2.5 | 4.5 | 10 | 0.420 | 0.924 | 142 |

(1) Pulsed (see 4.5.1).

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.5 for PIN construction example and 6.6 for a list of available PINs.

1.5.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: “JANTXV” and "JANS".

1.5.2 Radiation hardness assurance (RHA) designator. The RHA levels that are applicable for this specification sheet from lowest to highest for JANTXV and JANS quality levels are as follows: "M", "D", "P", "L", and "R".

1.5.3 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.5.3.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

1.5.3.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "7497", "7498", “7499” and "7561".

1.5.4 Suffix letters. The suffix letters "T2" are used on devices that are packaged in the TO-205AF package of figure 1.

1.5.5 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.



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* 1.6 Radiation features. The following radiation features are applicable for RHA devices supplied to this specification sheet.

* 1.6.1 Maximum total ionizing dose (TID). The maximum TID that RHA devices were tested to in accordance with method 1019 of MIL-STD-750 are as follows:

For device type 2N7497T2, AND2N7498T2:..... 100 krad(Si) 1/

* 1.6.2 Heavy ion irradiation SEE. SEE testing includes Single-Event Burnout (SEB) and Single-Event Gate Rupture (SEGR) tests at the specified linear energy transfer (LET) units. See 6.7 for the specific RHA SEE characterization testing details. The following characterization conditions were used for the testing to the requirements herein:

For device type 2N7497T2:
No SEB or SEGR were observed at surface LET (see table IV) ..... ≤ 84 MeV·cm$^2$/mg 2/
(In-situ Bias $V_{DS}$ = 130 V and $V_{GS}$ = 0 V)
(In-situ Bias $V_{DS}$ = 120 V and $V_{GS}$ = -5 V)
(In-situ Bias $V_{DS}$ = 30 V and $V_{GS}$ = - 10 V)

For device type 2N7498T2:
No SEB or SEGR were observed at surface LET (see table IV) ..... ≤ 84 MeV·cm$^2$/mg 2/
(In-situ Bias $V_{DS}$ = 200 V and $V_{GS}$ = -5V)
(In-situ Bias $V_{DS}$ = 150 V and $V_{GS}$ = -10 V)
(In-situ Bias $V_{DS}$ = 50 V and $V_{GS}$ = -15 V)
(In-situ Bias $V_{DS}$ = 25 V and $V_{GS}$ = -20 V)

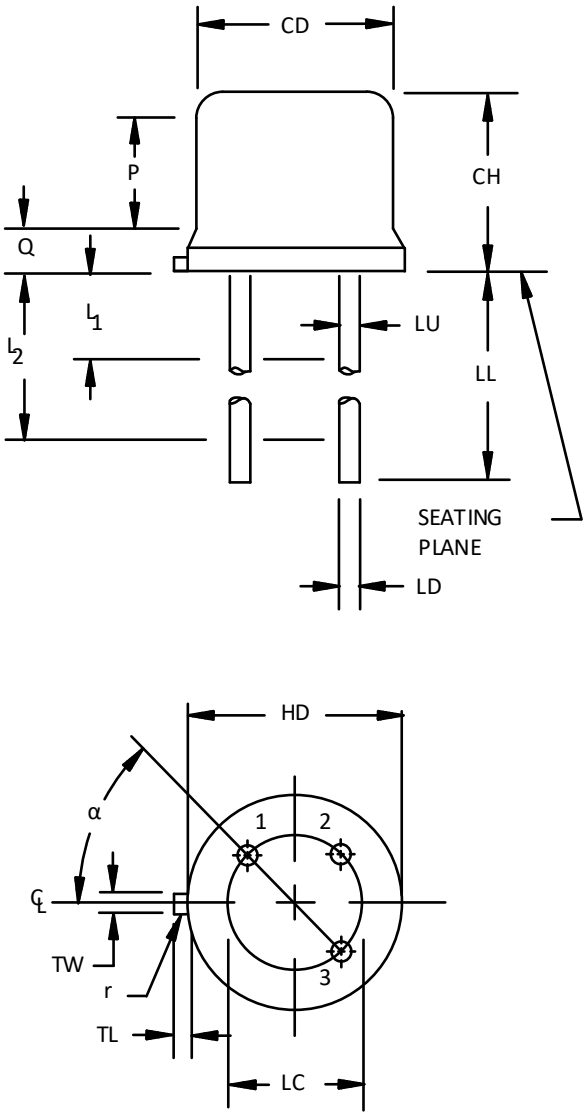
For device type 2N7499T2 ,2N7561T2:
No SEB or SEGR were observed at surface LET (see table IV) ..... ≤ 84 MeV·cm$^2$/mg 2/
(In-situ Bias $V_{DS}$ = 250 V and $V_{GS}$ = -5 V)
(In-situ Bias $V_{DS}$ = 225 V and $V_{GS}$ = -10 V)
(In-situ Bias $V_{DS}$ = 175 V and $V_{GS}$ = -15 V)
(In-situ Bias $V_{DS}$ = 50 V and $V_{GS}$ = -20 V)

* 1/ The manufacturers supplying these device types have performed characterization testing in accordance with method 1019 of MIL-STD-750. The radiation end point limits are guaranteed only to a maximum TID level of 100 krad(Si). For additional TID information contact the manufacturer.
* 2/ The manufacturers have performed SEE tests at Brookhaven National Lab Accelerator in accordance with method1080 of MIL-STD-750. For more information on SEE testing and results, customers should contact the qualified manufacturer.



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| Symbol (see note 3) | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| CD | .315 | .355 | 8.00 | 9.02 |
| CH | .160 | .180 | 4.06 | 4.57 |
| h | .009 | .041 | 0.23 | 1.04 |
| HD | .340 | .370 | 8.64 | 9.40 |
| LC | .200 BSC | | 5.08 BSC | |
| LD | .016 | .021 | 0.41 | 0.53 |
| LL | .500 | .750 | 12.70 | 19.05 |
| LU | .016 | .019 | 0.41 | 0.48 |
| $L_1$ | | .050 | | 1.27 |
| $L_2$ | .250 | | 6.35 | |
| P | .070 | | 1.78 | |
| Q | | .050 | | 1.27 |
| r | | .010 | | 0.25 |
| TL | .029 | .045 | 0.74 | 1.14 |
| TW | .028 | .034 | 0.71 | 0.86 |
| α | 45° BSC | | | |
| Term 1 | Source | | | |
| Term 2 | Gate | | | |
| Term 3 | Drain | | | |




NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Lead number 1 is the source, lead number 2 is the gate, and lead number 3 is the drain which is electrically connected to the case. Lead 4 is omitted from this outline.
4. Radius (r) applies to both inside corners of tab.
5. Dimension LU applies between L1 and L2. Dimension LD applies between L2 and LL minimum. Diameter is uncontrolled in L1 and beyond LL minimum.

FIGURE 1. Physical dimensions for TO-205AF.



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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

$I_{AS}$ ..... Rated avalanche current, nonrepetitive
nC ..... nano Coulomb.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1 (TO-205AF) herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Internal construction. Multiple chip construction shall not be permitted.



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3.5 Electrostatic discharge sensitive (ESDS). The devices covered by this specification sheet have been classified as ESDS. The devices shall be handled in accordance with the ESD program established to comply with the requirements of MIL-PRF-19500 to avoid damage due to the accumulation of static charge. The following handling practices shall be followed:

a. Devices should be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care should be exercised during test and troubleshooting to apply not more than maximum rated voltage to any lead.

* h. Gate must be terminated to source, $R \leq 100\ k\Omega$, whenever bias voltage is applied drain to source.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Marking. Marking shall be in accordance with MIL-PRF-19500. Lead finish designator shall not be marked on the device.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.2.1.1 Single event effects (SEE). SEE shall be performed at initial qualification and after process or design changes which may affect radiation hardness (see table III and table IV). Upon qualification, manufacturers shall provide the verification test conditions from section 5 of method 1080 of MIL-STD-750 that were used to qualify the device for inclusion into section 6 of the slash sheet. End-point measurements shall be in accordance with table II. SEE characterization data shall be made available upon request of the qualifying or acquiring activity.



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4.3 Screening (JANS and JANTXV). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen<br>(1) (2) | Measurements for<br>JANS | Measurements for<br>JANTXV |
|---|---|---|
| (3) | Gate stress test (see 4.3.1) | Gate stress test (see 4.3.1) |
| (3) | Method 3470 of MIL-STD-750, $E_{AS}$ test (see 4.3.2) | Method 3470 of MIL-STD-750, $E_{AS}$ test (see 4.3.2) |
| (3) 3c | Method 3161 of MIL-STD-750, thermal impedance (see 4.3.3) | Method 3161 of MIL-STD-750, thermal impedance (see 4.3.3) |
| 9 | Subgroup 2 of table I herein<br>$I_{DSS1}$, $I_{GSSF1}$, $I_{GSSR1}$, as a minimum | Not applicable |
| 10 | Method 1042 of MIL-STD-750, test condition B | Method 1042 of MIL-STD-750, test condition B |
| 11 | $I_{GSSF1}$, $I_{GSSR1}$, $I_{DSS1}$, $r_{DS(ON)1}$, $V_{GS(TH)1}$<br>Subgroup 2 of table I herein.<br>$\Delta I_{GSSF1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{DSS1}$ = ±10 μA dc or ±100 percent of initial value, whichever is greater. | $I_{GSSF1}$, $I_{GSSR1}$, $I_{DSS1}$, $r_{DS(ON)1}$, $V_{GS(TH)1}$<br>Subgroup 2 of table I herein. |
| 12 | Method 1042 of MIL-STD-750, test condition A | Method 1042 of MIL-STD-750, test condition A |
| 13 | Subgroups 2 and 3 of table I herein<br>$\Delta I_{GSSF1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{DSS1}$ = ±10 μA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta r_{DS(ON)1}$ = ±20 percent of initial value.<br>$\Delta V_{GS(TH)1}$ = ±20 percent of initial value. | Subgroup 2 of table I herein<br>$\Delta I_{GSSF1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{DSS1}$ = ±10 μA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta r_{DS(ON)1}$ = ±20 percent of initial value.<br>$\Delta V_{GS(TH)1}$ = ±20 percent of initial value. |

(1) At the end of the test program, $I_{GSSF1}$, $I_{GSSR1}$, and $I_{DSS1}$ are measured.
(2) An out-of-family program to characterize $I_{GSSF1}$, $I_{GSSR1}$, $I_{DSS1}$, and $V_{GS(th)1}$ shall be invoked.
* (3) Shall be performed anytime after temperature cycling, screen 3a. JANTXV levels do not need to be repeated in screening requirements.



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4.3.1 Gate stress test. Apply $V_{GS}$ = 24 V, minimum for t = 250 μS, minimum.

4.3.2 Single pulse avalanche energy ($E_{AS}$).

a. Peak current ..... $I_{AS} = I_{D1}$.

b. Inductance: ..... $\left[\frac{2E_{AS}}{(I_{D1})^2}\right]\left[\frac{V_{BR} - V_{DD}}{V_{BR}}\right]$ mH minimum.

c. Gate to source resistor ($R_{GS}$) ..... $25 \leq R_{GS} \leq 200\ \Omega$.

d. Supply voltage ..... $V_{DD}$ = 25 V dc, except $V_{DD}$ = 50 V dc for 2N7499T2.

e. Gate voltage ..... $V_{GS}$ = 12 V dc.

f. Initial case temperature ..... $T_C$ = +25°C +10°C, -5°C.

g. Number of pulses to be applied ..... 1 pulse minimum.

4.3.3 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3161 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{MD}$, $t_{SW}$, (and $V_H$ where appropriate). See table III, group E, subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JANTXV) of MIL-PRF-19500, and as follows.

4.4.2.1 Group B inspection, table E-VIA (JANS) of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B3 | 1051 | Test condition G, 100 cycles. |
| B3 | 2077 | Scanning Electron Microscope (SEM). |
| B4 | 1042 | Intermittent operation life, condition D. No heat sink or forced-air cooling on the device shall be permitted during the on cycle. $t_{on}$ = 30 seconds minimum. |
| B5 | 1042 | Accelerated steady-state gate bias, condition B, $V_{GS}$ = rated; $T_A$ = +175°C, t = 24 hours minimum; or $T_A$ = +150°C, t = 48 hours minimum. |
| B5 | 1042 | Accelerated steady-state reverse bias, condition A, $V_{DS}$ = rated; $T_A$ = +175°C, t = 120 hours minimum; or $T_A$ = +150°C, t = 240 hours minimum. |
| B5 | 2037 | Bond strength, test condition D. |



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4.4.2.2 Group B inspection, table E-VIB (JANTXV) of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B2 | 1051 | Test condition G, 25 cycles. |
| B3 | 1042 | Intermittent operation life, condition D. No heat sink or forced-air cooling on the device shall be permitted during the on cycle. $t_{on}$ = 30 seconds minimum. |
| B5 and B6 | | Not applicable. |

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500 and as follows.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Test condition E; weight = 8 ounces, 3 arcs of 90 degrees |
| C5 | 3161 | Thermal resistance, see 4.3.3, $R_{\theta JC(max)}$ = 5.0°C/W. |
| C6 | 1042 | Intermittent operation life, condition D. No heat sink or forced-air cooling on the device shall be permitted during the on cycle. $t_{on}$ = 30 seconds minimum. |

4.4.4 Group D inspection. Group D inspection shall be conducted in accordance with table E-VIII of MIL-PRF-19500 and table II herein.

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table III herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Condition | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical inspection | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Thermal impedance 2/ | 3161 | See 4.3.3 | $Z_{\theta JC}$ | | | °C/W |
| Breakdown voltage drain to source | 3407 | $V_{GS}$ = 0, $I_D$ = 1 mA dc, bias condition C | $V_{(BR)DSS}$ | | | |
| 2N7497T2 | | | | 130 | | V dc |
| 2N7498T2 | | | | 200 | | V dc |
| 2N7499T2 | | | | 250 | | V dc |
| 2N7561T2 | | | | 250 | | V dc |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = 1 mA dc | $V_{GS(TH)1}$ | 2.5 | 4.5 | V dc |
| Gate current | 3411 | $V_{GS}$ = +20 V dc, bias condition C, $V_{DS}$ = 0 | $I_{GSSF1}$ | | +100 | nA dc |
| Gate current | 3411 | $V_{GS}$ = -20 V dc, bias condition C, $V_{DS}$ = 0 | $I_{GSSR1}$ | | -100 | nA dc |
| Drain current | 3413 | $V_{GS}$ = 0, bias condition C, $V_{DS}$ = 80 percent of rated $V_{DS}$, | $I_{DSS1}$ | | 10 | µA dc |
| Static drain to source on-state resistance | 3421 | $V_{GS}$ = 12 V dc, condition A, pulsed (see 4.5.1), $I_D$ = $I_{D2}$ | $r_{DS(ON)1}$ | | | |
| 2N7497T2 | | | | | 0.100 | Ω |
| 2N7498T2 | | | | | 0.240 | Ω |
| 2N7499T2 | | | | | 0.420 | Ω |
| 2N7561T2 | | | | | 0.420 | Ω |
| Forward voltage | 4011 | $V_{GS}$ = 0, condition A, $I_D$ = $I_{D1}$ | $V_{SD}$ | | 1.5 | V dc |

See footnotes at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Condition | | Min | Max | |
| Subgroup 3 | | | | | | |
| High temperature operation | | $T_C = T_J = +125°C$ | | | | |
| Gate current | 3411 | $V_{GS} = \pm 20$ V dc, bias condition C, $V_{DS} = 0$ | $I_{GSS2}$ | | ±200 | nA dc |
| Drain current | 3413 | $V_{GS} = 0$, bias condition C, $V_{DS}$ = 80 percent of rated $V_{DS}$ | $I_{DSS2}$ | | 25 | µA dc |
| Static drain to source on-state resistance | 3421 | $V_{GS}$ = 12 V dc, condition A, pulsed (see 4.5.1), $I_D = I_{D2}$ | $r_{DS(ON)3}$ | | | |
| 2N7497T2 | | | | | 0.200 | Ω |
| 2N7498T2 | | | | | 0.480 | Ω |
| 2N7499T2 | | | | | 0.840 | Ω |
| 2N7561T2 | | | | | 0.840 | Ω |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = 1 mA dc | $V_{GS(TH)2}$ | 1.5 | | V dc |
| Low temperature operation | | $T_C = T_J = -55°C$ | | | | |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS(TH)3}$, $I_D$ = 1 mA dc | $V_{GS(TH)3}$ | | 5.5 | V dc |
| Subgroup 4 | | | | | | |
| Forward transconductance | 3475 | $I_D = I_{D2}$, $V_{DD}$ = 15 V dc (see 4.5.1) | $g_{FS}$ | | | |
| 2N7497T2 | | | | 4.8 | | S |
| 2N7498T2 | | | | 4.2 | | S |
| 2N7499T2 | | | | 4.0 | | S |
| 2N7561T2 | | | | 4.0 | | S |
| Switching time test | 3472 | $I_D = I_{D1}$, $V_{GS}$ = 12 V dc, $R_G$ = 7.5 Ω, $V_{DD}$ = 50 percent of rated $V_{DS}$ | | | | |
| Turn-on delay time | | | $t_{D(on)}$ | | 25 | ns |
| Rise time | | | $t_r$ | | 100 | ns |
| Turn-off delay time | | | $t_{D(off)}$ | | 35 | ns |
| Fall time | | | $t_f$ | | 40 | ns |

See footnotes at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Condition | | Min | Max | |
| Subgroup 5 | | | | | | |
| Safe operating area test (high voltage) | 3474 | See figure 4, $t_p$ = 10 ms min. $V_{DS}$ = 80 percent of max. rated $V_{DS}$ | | | | |
| Electrical measurements | | See table I, subgroup 2 | | | | |
| Subgroup 6 | | | | | | |
| Not applicable | | | | | | |
| Subgroup 7 | | | | | | |
| Gate charge | 3471 | Condition B, $I_D = I_{D1}$, $V_{GS}$ = 12 V dc; $V_{DD}$ = 50 percent of rated $V_{DS}$ | | | | |
| On-state gate charge | | | $Q_{G(ON)}$ | | | |
| 2N7497T2 | | | | | 48 | nC |
| 2N7498T2 | | | | | 47 | nC |
| 2N7499T2 | | | | | 28 | nC |
| 2N7561T2 | | | | | 32 | nC |
| Gate to source charge | | | $Q_{GS}$ | | | |
| 2N7497T2 | | | | | 16 | nC |
| 2N7498T2 | | | | | 12 | nC |
| 2N7499T2 | | | | | 7.4 | nC |
| 2N7561T2 | | | | | 11 | nC |
| Gate to drain charge | | | $Q_{GD}$ | | | |
| 2N7497T2 | | | | | 18 | nC |
| 2N7498T2 | | | | | 16 | nC |
| 2N7499T2 | | | | | 12 | nC |
| 2N7561T2 | | | | | 16 | nC |
| Reverse recovery time | 3473 | Condition A, di/dt = -100 A/µs, $V_{DD} \leq 50$ V, $I_D = I_{D1}$ | $t_{rr}$ | | | |
| 2N7497T2 | | | | | 250 | ns |
| 2N7498T2 | | | | | 274 | ns |
| 2N7499T2 | | | | | 287 | ns |
| 2N7561T2 | | | | | 287 | ns |

1/ For sampling plan, see MIL-PRF-19500.
2/ For end-point measurements, this test is required for the following subgroups (not intended for 4.3, screen 9 or 11):
Group B, subgroups 2 and 3 (JANTXV).
Group B, subgroups 3 and 4 (JANS).
Group C, subgroups 2 and 6.
Group E, subgroup 1.



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TABLE II. Group D inspection.

| Inspection 1/ 2/ 3/ | MIL-STD-750 | | Symbol | Pre-irradiation limits | | Post-irradiation limits | | Unit |
|---|---|---|---|---|---|---|---|---|
| | | | | R | | R | | |
| | Method | Conditions | | Min | Max | Min | Max | |
| Subgroup 1 | | | | | | | | |
| Not applicable | | | | | | | | |
| Subgroup 2 | | $T_C$ = + 25°C | | | | | | |
| Steady-state total dose irradiation ($V_{GS}$ bias) 4/ | 1019 | $V_{GS}$ = 12 V; $V_{DS}$ = 0 | | | | | | |
| Steady-state total dose irradiation ($V_{DS}$ bias) 4/ | 1019 | $V_{GS}$ = 0; $V_{DS}$ = 80 percent of rated $V_{DS}$ (preirradiation) | | | | | | |
| End-point electricals | | | | | | | | |
| Breakdown voltage, drain to source | 3407 | $V_{GS}$ = 0; $I_D$ = 1 mA; bias condition C | $V_{(BR)DSS}$ | | | | | |
| 2N7497T2 | | | | 130 | | 130 | | V dc |
| 2N7498T2 | | | | 200 | | 200 | | V dc |
| 2N7499T2, 2N7561T2 | | | | 250 | | 250 | | V dc |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$; $I_D$ = 1 mA | $V_{GS(th)1}$ | 2.5 | 4.5 | 2.0 | 4.5 | V dc |
| Gate current | 3411 | $V_{GS}$ = +20 V; $V_{DS}$ = 0 bias condition C | $I_{GSSF1}$ | | 100 | | 100 | nA dc |
| Gate current | 3411 | $V_{GS}$ = -20 V; $V_{DS}$ = 0 bias condition C | $I_{GSSR1}$ | | -100 | | -100 | nA dc |
| Drain current | 3413 | $V_{GS}$ = 0; $V_{DS}$ = 80 percent of rated $V_{DS}$ (preirradiation) bias condition C | $I_{DSS}$ | | 10 | | 10 | µA dc |
| Static drain to source on-state voltage | 3405 | $V_{GS}$ = 12 V; $I_D$ = $I_{D2}$; condition A, pulsed (see 4.5.1) | $V_{DS(on)}$ | | | | | |
| 2N7497T2 | | | | | 0.533 | | 0.533 | V dc |
| 2N7498T2 | | | | | 0.955 | | 0.955 | V dc |
| 2N7499T2, 2N7561T2 | | | | | 1.326 | | 1.326 | V dc |
| Forward voltage source drain diode | 4011 | $V_{GS}$ = 0; $I_D$ = $I_{D1}$ bias condition A | $V_{SD}$ | | 1.5 | | 1.5 | V dc |

1/ For sampling plan see MIL-PRF-19500.
2/ Group D qualification may be performed prior to lot formation. Wafers qualified to these group D QCI requirements may be used for any other specification sheet utilizing the same die design.
3/ At the manufacturer's option, group D samples need not be subjected to the screening tests, and may be assembled in it's qualified package or in any qualified package that the manufacturer has data to correlate the performance to the designated package.
4/ Separate samples shall be pulled for each bias. For additional TID information contact the manufacturer.



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TABLE III. Group E inspection (all quality levels) for qualification or re-qualification only.

| Inspection | MIL-STD-750 | | Qualification and large lot |
|---|---|---|---|
| | Method | Conditions | Quality conformance inspection |
| Subgroup 1 | | | 45 devices c = 0 |
| Temperature cycling | 1051 | Test condition G, 500 cycles | |
| Hermetic seal<br>Fine leak<br>Gross leak | 1071 | | |
| Electrical measurements | | Table I, subgroup 2 herein. | |
| Subgroup 2 1/ | | | 45 devices c = 0 |
| Steady-state reverse bias | 1042 | Condition A, 1,000 hours. | |
| Electrical measurements | | Table I, subgroup 2 herein. | |
| Steady-state gate bias | 1042 | Condition B, 1,000 hours. | |
| Electrical measurements | | Table I, subgroup 2 herein. | |
| Subgroup 4 | | | Sample size N/A |
| Thermal impedance curves | | See MIL-PRF-19500. | |
| Subgroup 5 | | | 15 devices c = 0 |
| Barometric pressure (2N7498T2, 2N7499T2 & 2N7561T2 only) | 1001 | Test condition C. $V_{DS}$ = 250 V; $I_{(ISO)}$ < 0.25 mA. | |
| Subgroup 10 | | | 22 devices c = 0 |
| Commutating diode for safe operating area test procedure for measuring dv/dt during reverse recovery of power MOSFET transistors or insulated gate bipolar transistors | 3476 | Test conditions shall be derived by the manufacturer. | |
| Subgroup 11 | | | |
| SEE 2/ 3/ | 1080 | See method 1080 of MIL-STD-750. | 3 devices |

1/ A separate sample for each test shall be pulled.
2/ Group E qualification of SEE testing may be performed prior to lot formation. Qualification may be extended to other specification sheets utilizing the same structurally identical die design.
3/ Device qualification to a higher level LET is sufficient to qualify all lower level LETs.



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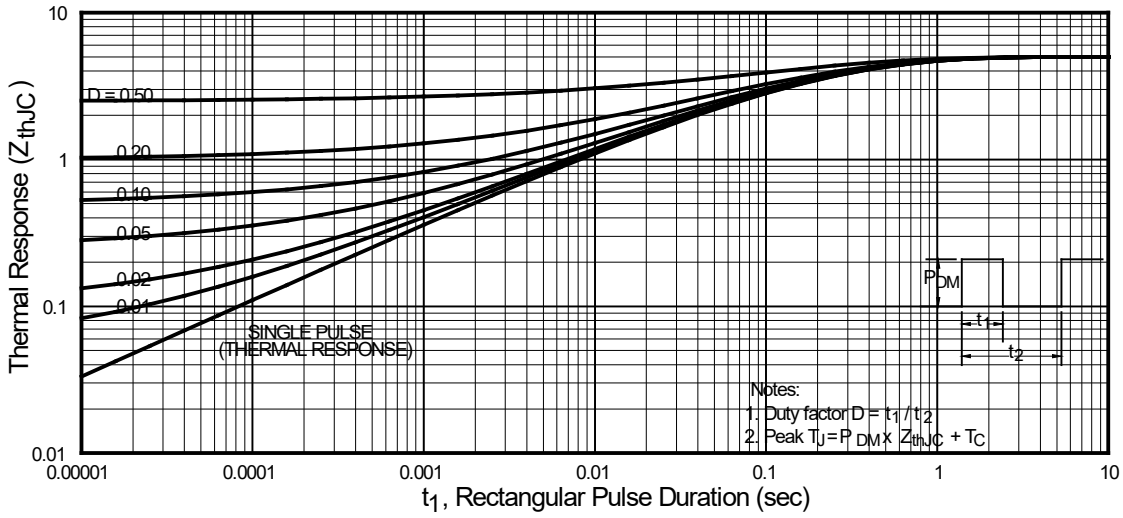
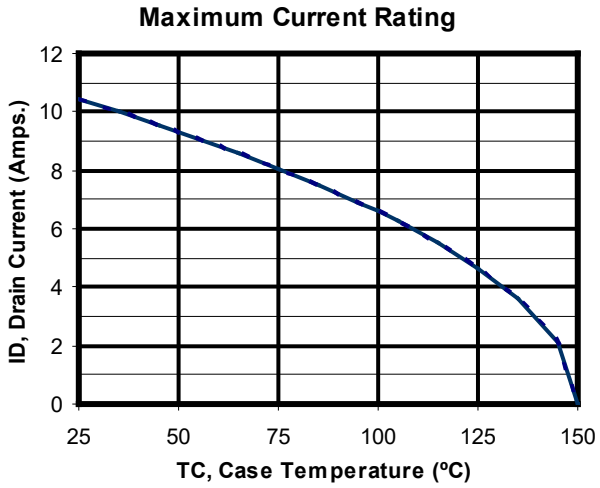


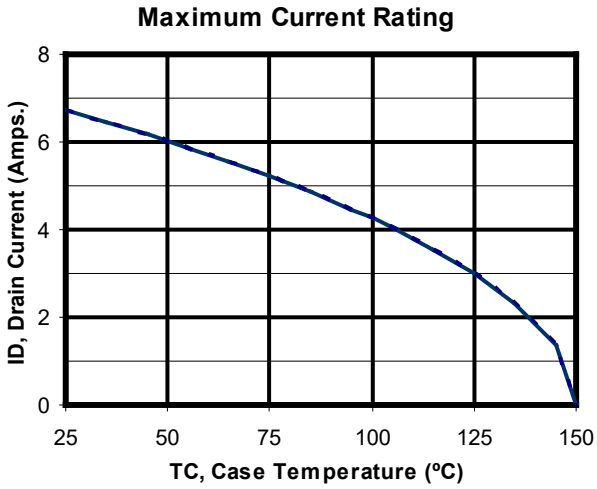

FIGURE 2. Thermal impedance curve.



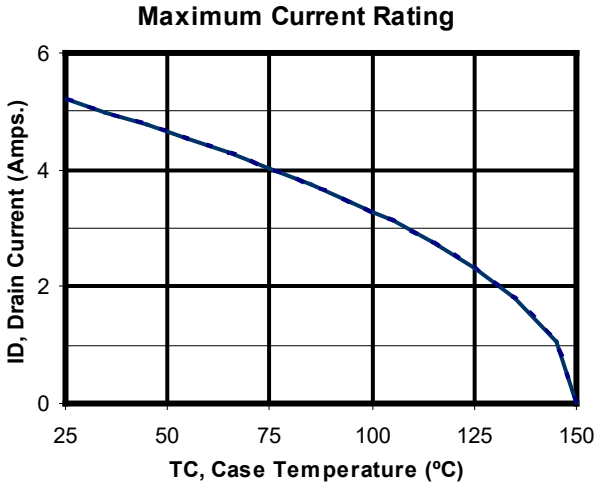
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2N7497T2




2N7498T2

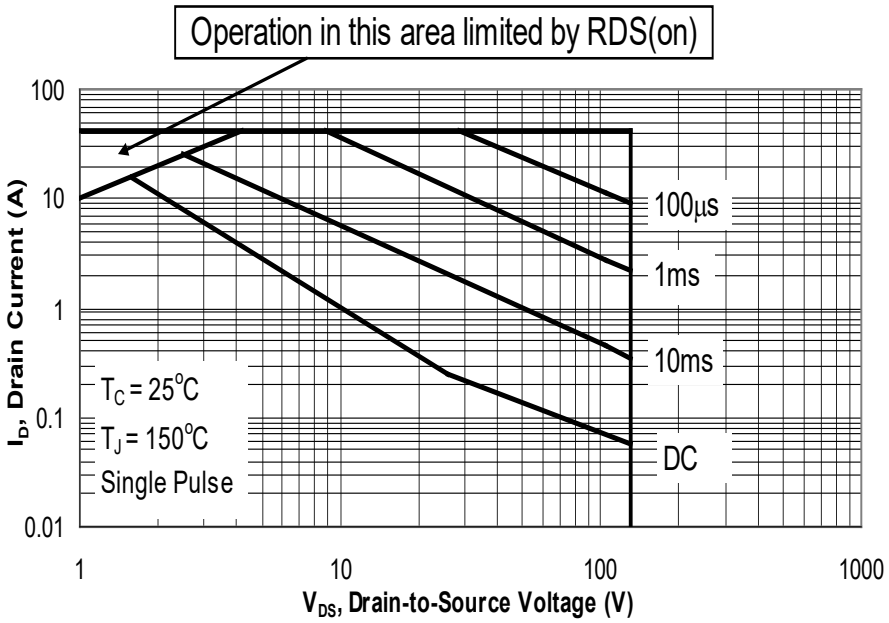



2N7499T2 and 2N7561T2

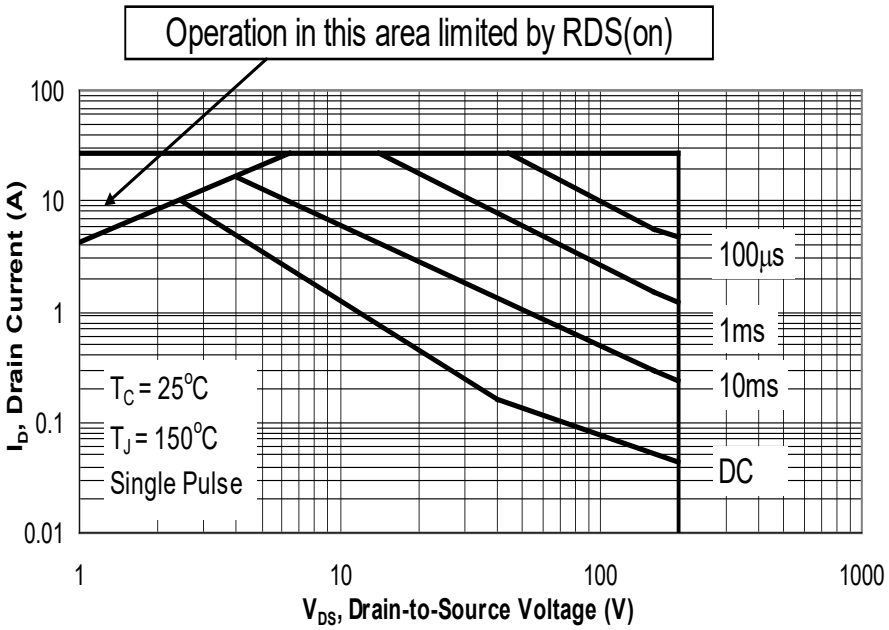
FIGURE 3. Maximum drain current vs case temperature graphs.



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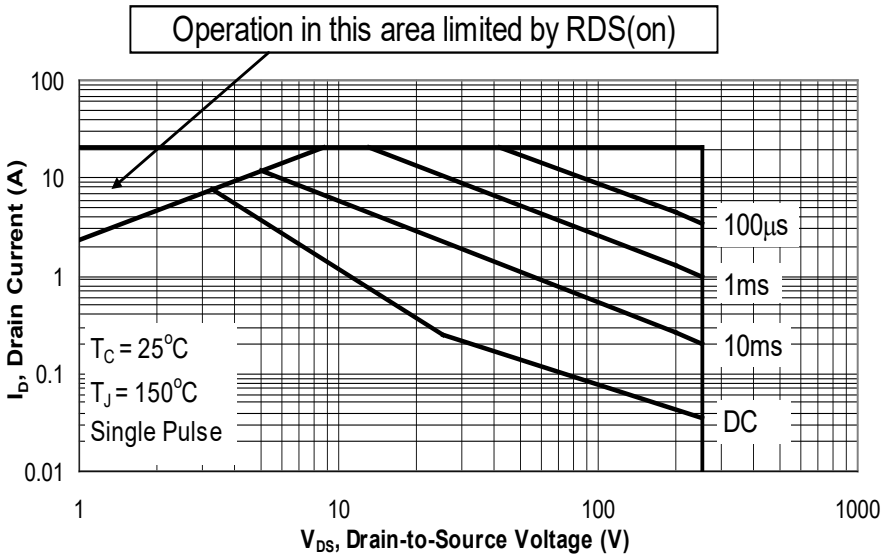



2N7498T2

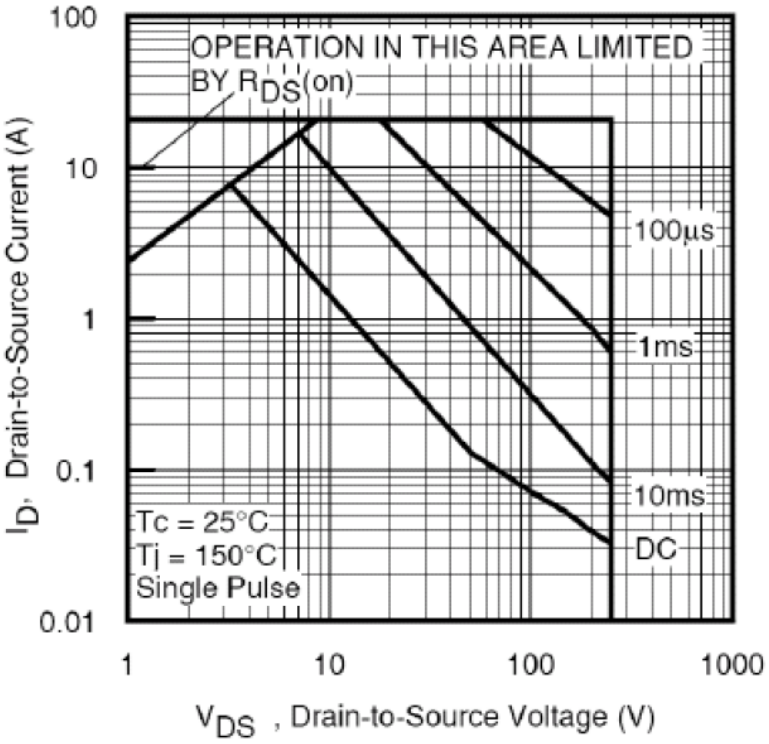
FIGURE 4. Safe operating area graphs.



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2N7499T2




2N7561T2

FIGURE 4. Safe operating area graphs – Continued.



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete Part or Identifying Number (PIN), see 1.5 and 6.5.

e. For acquisition of RHA designated devices, table II, subgroup 1 testing of group D herein is optional. If subgroup 1 is desired, it should be specified in the contract.

f. If SEE testing data is desired, it should be specified in the contract or order (see 4.2.1.1).

g. If specific SEE characterization conditions are desired (see section 6.7 and table IV), manufacturer's CAGE code should be specified in the contract or order.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.



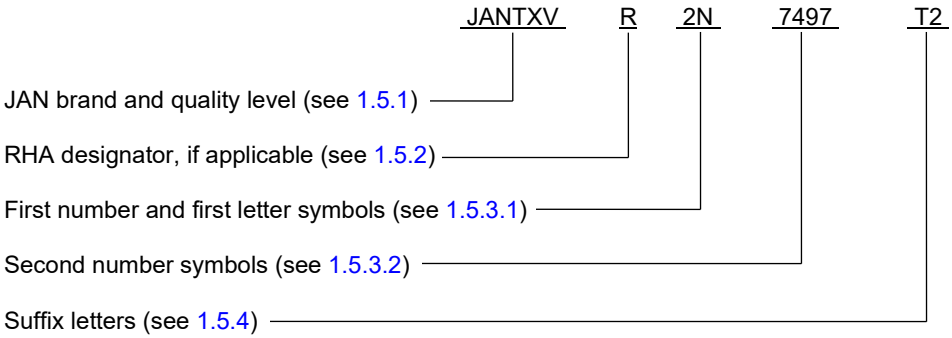
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6.4 Cross-reference list. The following table shows the generic P/N and its associated military P/N (without JAN and RHA prefix).

| Generic P/N | Military P/N |
|---|---|
| IRHF57133SE | 2N7497T2 |
| IRHF57230SE | 2N7498T2 |
| IRHF57234SE | 2N7499T2<br>2N7561T2 |

6.5 PIN construction example.

6.5.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.




6.6 List of PINs. The following is a list of possible PINs available on this specification sheet.

| PINs for devices of the "TXV" quality level | PINs for devices of the "TXV" quality level with RHA (1) | PINs for devices of the "S" quality level | PINs for devices of the "S" quality level with RHA (1) |
|---|---|---|---|
| JANTXV2N7497T2 | JANTXV#2N7497T2 | JANS2N7497T2 | JANS#2N7497T2 |
| JANTXV2N7498T2 | JANTXV#2N7498T2 | JANS2N7498T2 | JANS#2N7498T2 |
| JANTXV2N7499T2 | JANTXV#2N7499T2 | JANS2N7499T2 | JANS#2N7499T2 |
| JANTXV2N7561T2 | JANTXV#2N7561T2 | JANS2N7561T2 | JANS#2N7561T2 |

(1) The number sign (#) represent one of five RHA designators available (M, D, P, L, or R).

6.7 Application data.

6.7.1 Manufacturer specific irradiation data. Each manufacturer qualified to this slash sheet has characterized its devices to the requirements of MIL-STD-750 method 1080 and as specified herein. Since each manufacturer's characterization conditions can be different and can vary by the version of method 1080 qualified to, the MIL-STD-750 method 1080 revision version date and conditions used by each manufacturer for characterization have been listed here (see table IV) for information only. SEE conditions and figures listed in section 6 are current as of the date of this specification sheet, please contact the manufacturer for the most recent conditions.



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TABLE IV. Manufacturers characterization conditions.

| Manufactures CAGE | Inspection | MIL-STD-750 | | Sample plan |
|---|---|---|---|---|
| | | Method | Conditions | |
| 69210 (Applicable to devices with a date code of 16 June 1998 and older) | SEE 1/ | 1080 | See MIL-STD-750 method 1080.0 dated 20 November 2006. See figure 5 | |
| | Electrical measurements | | $I_{GSSF1}$, $I_{GSSR1}$, and $I_{DSS1}$ in accordance with table I, subgroup 2 | 3 devices |
| | SEE irradiation: | | Fluence = 3E5 ±20 percent ions/cm$^2$<br>Flux = 2E3 to 2E4 ions/cm$^2$/sec, temperature = 25 ±5°C<br><br>Surface LET = 38 MeV-cm2/mg ± 5%<br>Range = 38 µm ±7.5%, Energy = 300 MeV ±7.5% | |
| | 2N7497T2 | | In-situ bias conditions: $V_{DS}$ = 130 V and $V_{GS}$ = -20 V<br>(typical 3.75 MeV/nucleon at Texas A & M Cyclotron) | |
| | 2N7498T2 | | In-situ bias conditions: $V_{DS}$ = 200 V and $V_{GS}$ = -20 V<br>(nominal 3.86 MeV/nucleon at Brookhaven National Lab Accelerator) | |
| | 2N7499T2 & 2N7561T2 | | In-situ bias conditions: $V_{DS}$ = 250 V and $V_{GS}$ = -20 V<br>(nominal 3.86 MeV/nucleon at Brookhaven National Lab Accelerator)<br><br>Surface LET = 61 MeV-cm2/mg ±5%<br>Range = 31 µm ±10%, Energy = 330 MeV ±7.5% | |
| | 2N7497T2 | | In-situ bias conditions: $V_{DS}$ = 130 V and $V_{GS}$ = -10 V<br>$V_{DS}$ = 100 V and $V_{GS}$ = -15 V<br>$V_{DS}$ = 50 V and $V_{GS}$ = -20 V<br>(typical 2.70 MeV/nucleon at Texas A & M Cyclotron) | |
| | 2N7498T2 | | In-situ bias conditions: $V_{DS}$ = 200 V and $V_{GS}$ = -10 V<br>$V_{DS}$ = 185 V and $V_{GS}$ = -15 V<br>$V_{DS}$ = 120 V and $V_{GS}$ = -20 V<br>(nominal 2.92 MeV/nucleon at Brookhaven National Lab Accelerator) | |
| | 2N7499T2 & 2N7561T2 | | In-situ bias conditions: $V_{DS}$ = 250 V and $V_{GS}$ = -15 V<br>$V_{DS}$ = 240 V and $V_{GS}$ = -20 V<br>(nominal 2.92 MeV/nucleon at Brookhaven National Lab Accelerator) | |

See footnotes at end of table.



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TABLE IV. Manufacturers characterization conditions - Continued.

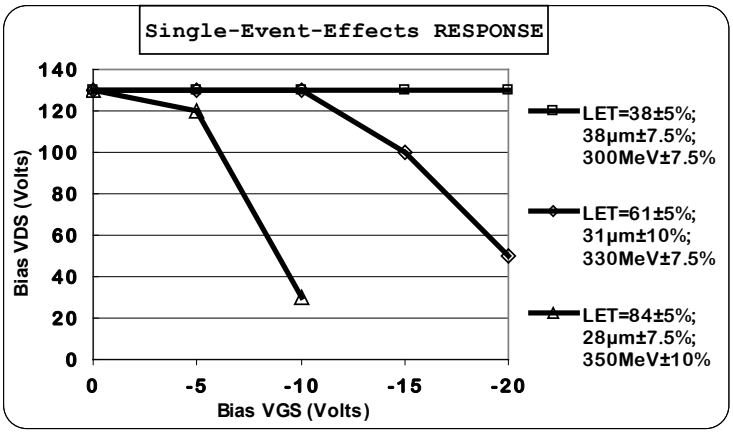
| Manufactures CAGE | Inspection | MIL-STD-750 | | Sample plan |
|---|---|---|---|---|
| | | Method | Conditions | |
| | | | Surface LET = 84 MeV-cm2/mg ± 5%<br>Range = 28 µm ±7.5%, Energy = 350 MeV ±10% | 3 devices |
| | 2N7497T2 | | In-situ bias conditions: $V_{DS}$ = 130 V and $V_{GS}$ = 0 V<br>$V_{DS}$ = 120 V and $V_{GS}$ = -5 V<br>$V_{DS}$ = 30 V and $V_{GS}$ = - 10 V<br>(typical 1.89 MeV/nucleon at Texas A & M Cyclotron) | |
| | 2N7498T2 | | In-situ bias conditions: $V_{DS}$ = 200 V and $V_{GS}$ = -5V<br>$V_{DS}$ = 150 V and $V_{GS}$ = -10 V<br>$V_{DS}$ = 50 V and $V_{GS}$ = -15 V<br>$V_{DS}$ = 25 V and $V_{GS}$ = -20 V<br>(nominal 1.98 MeV/nucleon at Brookhaven National Lab Accelerator) | |
| | 2N7499T2 & 2N7561T2 | | In-situ bias conditions: $V_{DS}$ = 250 V and $V_{GS}$ = -5 V<br>$V_{DS}$ = 225 V and $V_{GS}$ = -10 V<br>$V_{DS}$ = 175 V and $V_{GS}$ = -15 V<br>$V_{DS}$ = 50 V and $V_{GS}$ = -20 V<br>(nominal 1.98 MeV/nucleon at Brookhaven National Lab Accelerator) | |
| | Electrical measurements | | $I_{GSSF1}$, $I_{GSSR1}$, and $I_{DSS1}$ in accordance with table I, subgroup 2 | |
| Upon qualification, all manufacturers should provide the verification test conditions to be added to this table. | | | | |

1/ $I_{GSSF1}$, $I_{GSSR1}$, and $I_{DSS1}$ was examined before and following SEE irradiation to determine acceptability for each bias condition. Other test conditions in accordance with table I, subgroup 2, may be performed at the manufacturer's option.

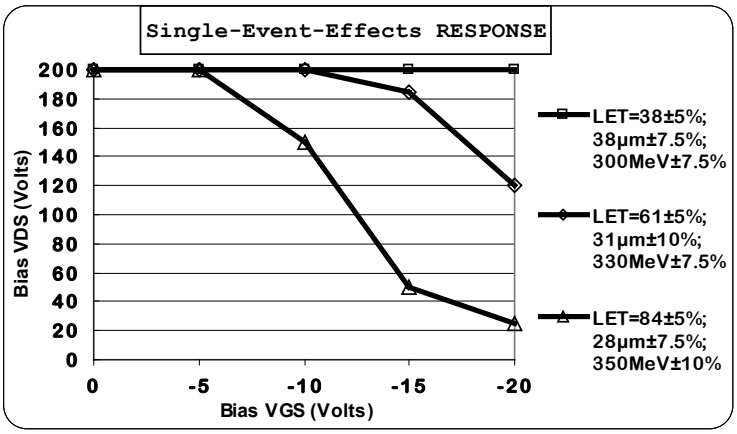


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2N7497T2




2N7498T2




2N7499T2 and 2N7561T2

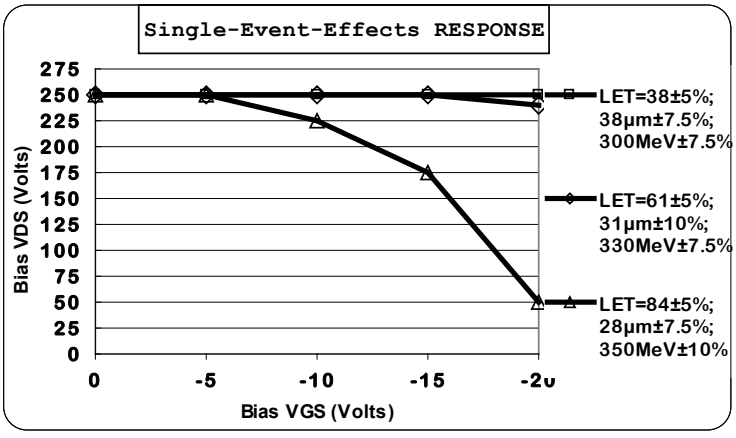



FIGURE 5. Typical SEE safe operating area graphs.



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6.8 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
* Navy - SH
Air Force - 85
NASA - NA
DLA - CC

Review activities:
Army - MI
Air Force - 19

Preparing activity:
DLA - CC

(Project 5961-2025-047)

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